MODULE-1

Introduction to Mechanical Engineering (ME131)

Module 1 (10 hrs)

Statics of Particles: Force, Classification & representation, Force as a vector, Parallelogram law, Resolution and composition of forces, Principle of superposition, and Transmissibility of forces.

Statics of Rigid bodies: Equilibrium of coplanar force system, Free body diagrams, Determination of reactions, Equilibrium of a body under three forces, Lami's theorem. Moment of a force about a point and an axis, Moment of coplanar force system, Varignon's theorem.

Friction: Introduction, Limiting friction, Angle of friction, Laws of dry friction, Friction in inclined planes, Angle of repose, Cone of friction.

1.1. Introduction

The state of rest and state of motion of the bodies under the action of different forces has engaged the attention of philosophers, mathematicians, and scientists for many centuries. The branch of physical science which describes and predicts the conditions of rest or the state of motion of bodies under the action of forces is termed **Mechanics.** Starting from the analysis of rigid bodies under gravitational force and simply applied forces the mechanics has grown to the analysis of robotics, aircraft, and spacecraft under dynamic forces, atmospheric forces, temperature forces, etc. Archimedes (287–212 BC), Galileo (1564–1642), Sir Isaac Newton (1642–1727), and Einstein (1878–1955) have contributed a lot to the development of mechanics. Contributions by Varignon, Euler, D. Alembert are also substantial. The mechanics developed by these researchers may be grouped as

(*i*) Classical mechanics/Newtonian mechanics

(*ii*) Relativistic mechanics

(*iii*) Quantum mechanics/Wave mechanics.

Sir Issac Newton, the principal architect of mechanics, consolidated the philosophy and experimental findings developed around the state of rest and motion of the bodies and put forth them in the form of three laws of motion as well as the law of gravitation. The mechanics based on these laws is called **Classical mechanics** or **Newtonian mechanics.** Albert Einstein proved that Newtonian mechanics fails to explain the behavior of high-speed (speed of light) bodies. He put forth the theory of **Relativistic Mechanics.** Schrödinger (1887–1961) and Broglie (1892–1965) showed that Newtonian mechanics fails to explain the behavior of particles when atomic distances are concerned. They put forth the theory of **Quantum Mechanics.** Engineers are keen to use the laws of mechanics to actual field problems. Application of laws of mechanics to field problems is termed as **Engineering Mechanics.** For all the problems between atomic distances to high-speed distances Classical/Newtonian mechanics has stood the test of time and hence that is the mechanics used by engineers. Hence in this text, classical mechanics is used for the analysis of engineering problems.

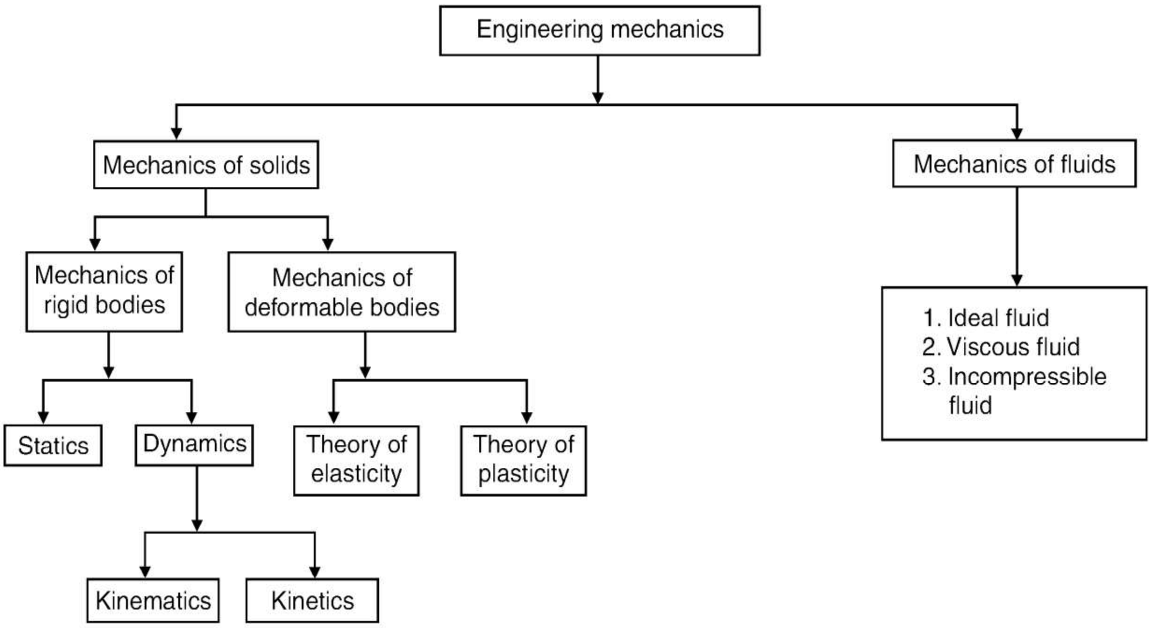
1.2. Classification of Engineering Mechanics

Depending upon the body to which the mechanics is applied, the engineering mechanics is classified as

(*a*) Mechanics of Solids, and

(*b*) Mechanics of Fluids.

The classification of mechanics is summarized below in a flow chart.



The solid mechanics is further classified as mechanics of rigid bodies and mechanics of deformable bodies. The body that will not deform or the body in which deformation can be neglected in the analysis are called **Rigid bodies**. The mechanics of the rigid bodies dealing with the bodies at rest is termed **Statics** and those dealing with bodies in motion is called **Dynamics**. The dynamics dealing with the problems without referring to the forces causing the motion of the body is termed as **Kinematics** and if it deals with the forces causing motion also, is called **Kinetics.** If the internal stresses developed in a body are to be studied, the deformation of the body should be considered. This field of mechanics is called **Mechanics of Deformable Bodies/ Strength of Materials/Solid Mechanics**. This field may be further divided into **Theory of Elasticity** and **Theory of Plasticity.** Liquid and gases deform continuously with application of very small shear forces. Such materials are called **Fluids.**

The mechanics dealing with behavior of such materials is called **Fluid Mechanics.** Mechanics of ideal fluids, mechanics of viscous fluid, and mechanics of incompressible fluids are further classified in this area.

1.3. Basic Terminologies

The following are the terms basic to study mechanics, which should be understood clearly:

Mass

The quantity of the matter possessed by a body is called mass. The mass of a body will not change unless the body is damaged and part of it is physically separated. When a body is taken out in a spacecraft, the mass will not change but its weight may change due to change in gravitational force. Even the body may become weightless when gravitational force vanishes but the mass remains the same.

Time

Time is the measure of succession of events. The successive event selected is the rotation of earth about its own axis and this is called a day. To have convenient units for various activities, a day is divided into 24 hours, an hour into 60 minutes and a minute into 60 seconds. Clocks are the instruments developed to measure time. To overcome difficulties due to irregularities in the earth's rotation, the unit of time is taken as second, which is defined as the duration of 9192631770 period of radiation of the cesium-133 atom.

Space

The geometric region in which study of body is involved is called **space.** A point in the space may be referred to with respect to a predetermined point by a set of linear and angular measurements. The reference point is called the origin and the set of measurements as '*coordinates*'. If coordinates involve only in mutually perpendicular directions, they are known as *Cartesian coordinates.* If the coordinates involve angles and distances, it is termed as *polar coordinate system.*

Length

It is a concept to measure linear distances. The diameter of a cylinder may be 300 mm, and the height of a building may be 15 m. Actually, the meter is the unit of length. However, depending upon the sizes involved micro, milli or kilo meter units are used for measurement.

A meter is defined as length of the standard bar of platinum-iridium kept at the International Bureau of Weights and Measures. To overcome difficulties of accessibility and reproduction, now meter is defined as 1690763.73 wavelength of krypton-86 atom.

Displacement

Displacement is defined as the distance moved by a body/particle in the specified direction.

Velocity

The rate of change of displacement with respect to time is defined as velocity.

Acceleration

Acceleration is the rate of change of velocity with respect to time.

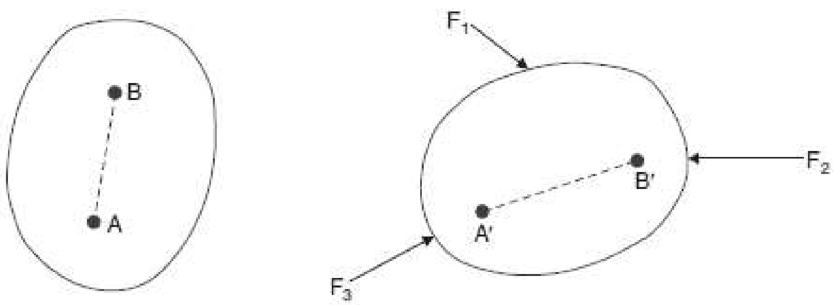
Momentum

The product of mass and velocity is called momentum. Thus Momentum = Mass × Velocity

Continuum

A body consists of several matters. It is a well-known fact that each particle can be subdivided into molecules, atoms and electrons. It is not possible to solve any engineering problem by treating a body as a conglomeration of such discrete particles. The body is assumed to consist of a continuous distribution of matter. In other words, the body is treated as **continuum.**

Rigid Body



(a) (b)

A body is said to be rigid if the relative positions of any two particles in it do not change under the action of the forces. In above Fig. (*a*) points A and B are the original positions in a body. After application of a system of forces $F1$, $F2$, and $F3$, the body takes the position as shown in above Fig. (*b*). A' and B' are the new positions of A and B. If the body is treated as rigid, the relative position of $A'B'$ and AB are the same *i.e.,* $A'B' = AB$.

Particle

A particle may be defined as an object which has only mass and no size. Such a body cannot exist theoretically. However, in dealing with problems involving distances considerably larger compared to the size of the body, the body may be treated as particle, without sacrificing accuracy. Examples of such situations are

- A bomber airplane is a particle for a gunner operating from the ground.
- A ship in mid sea is a particle in the study of its relative motion from a control tower.
- In the study of movement of the earth in celestial sphere, earth is treated as a particle.

Scalar quantities

These are the quantities characterized by only magnitude, such as mass, temperature.

Vector quantities

These are the quantities characterized by magnitude and direction, such as velocity, force. We will indicate vector quantities by boldfaced letters.

Free vector

It is a vector, whose action is not confined to or associated with a unique line in space. Only, the direction and magnitude of the vector remains fixed.

Unit Vector

It is a vector, whose magnitude is unity.

Equal Vectors

These are vectors, which are parallel to each other & same in magnitude.

Like Vector

These are vectors, which are parallel to each other, but not same in magnitude.

1.4. Laws of Mechanics

The following are the fundamental laws of mechanics:

- Newton’s first law of motion
- Newton’s second law of motion
- Newton’s third law of motion

- Newton's law of gravitation
- Law of transmissibility of forces, and
- Parallelogram law of forces.

Newton's First Law of Motion

It states that everybody continues in its state of rest or of uniform motion in a straight line unless it is compelled by external agency acting on it. This leads to the definition of force as the external agency which changes or tends to change the state of rest or uniform linear motion of the body.

Newton's Second Law of Motion

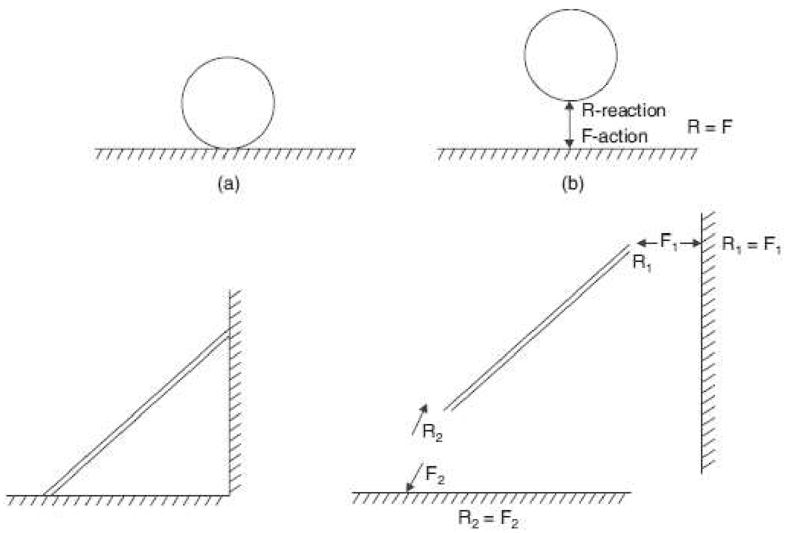
It states that the rate of change of momentum of a body is directly proportional to the impressed force and it takes place in the direction of the force acting on it. Thus according to this law, Force is directly proportional to the rate of change of momentum. But momentum = mass × velocity

As mass do not change, Force ∝ mass × rate of change of velocity

i.e., Force ∝ mass × acceleration

$F \propto m \times a$

Newton's Third Law of Motion



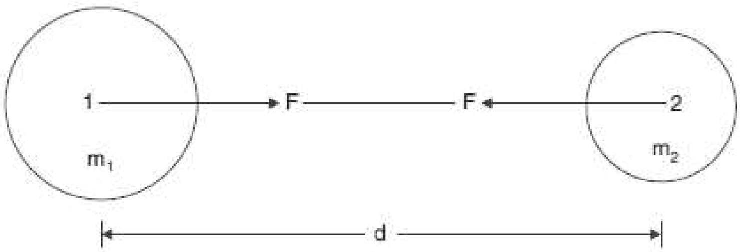
It states that for every action there is an equal and opposite reaction. Consider the two bodies in contact with each other. Let one body applies a force F on another. According to this law the second body develops a reactive force R which is equal in magnitude to force F and acts in the line same as F but in the opposite direction. Above Figures shows the action and the reaction forces are associated with the ball and the ladder respectively.

Newton's Law of Gravitation

Everybody attracts the other body. The force of attraction between any two bodies is directly proportional to their masses and inversely proportional to the square of the distance between them. According to this law the force of attraction between the bodies of mass m_1 and mass m_2 at a distance d as shown in Fig. 1.5 is

$$F = G\frac{m_1 m_2}{d^2}$$

Where, G is the constant of proportionality and is known as constant of gravitation.



1.5. Force

It is defined as any action that tends to change the state of rest or motion of a body to which it is applied. Force is the action of one body on another. A force tends to move the body in the direction of its action. The action of a force is characterized by its magnitude, the direction of its action, and its points of application. i.e. A force is completely defined when its

(i) Magnitude,
(ii) Direction,
(iii) Point of application
(iv) Line of action are well defined.

A force may be of following types. (i) Applied force.
(ii) Non-Applied force.

Non-Applied force may further categorize as (i) Self-weight.

(ii) Reaction.

System of Forces: When two or more than two forces are acting upon a body, they are called to form a system of forces or force system.

The Force system can be classified as (i) Co-planner force system.

(ii) Non-Co planner force system.

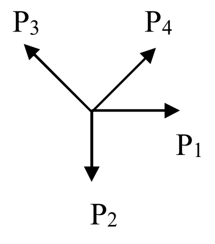
System of forces

- Coplanar forces
  - Concurrent
  - Parallel
    - Like
    - Unlike
  - Non-current Or Non-parallel
  - collinear
- Non-coplanar or spatial forces
  - concurrent
  - parallel
    - Like
    - Unlike
  - Non-concurrent or Non-parallel

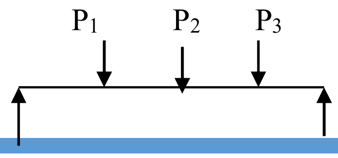
Example in graphical form:

(1) Coplanar collinear forces:

(2) Coplanar concurrent forces:

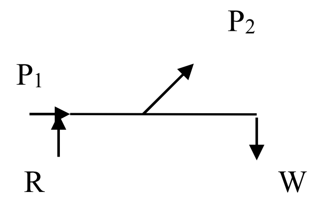


(3) Coplanar parallel forces:

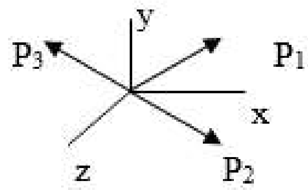


R_1 R_2

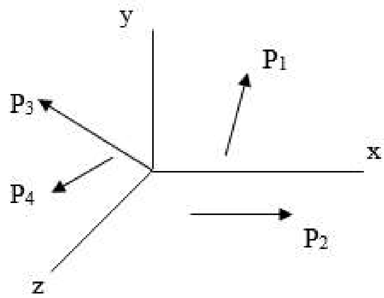
(4) Coplanar non-concurrent forces:



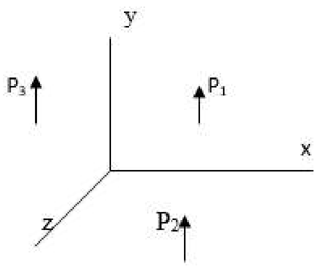
(5) Non-coplanar concurrent forces:



(6) Non-coplanar non-concurrent forces:



(7) Non-coplanar parallel forces:



1.6. Resultant Force

If a number of forces F_1, F_2, and F_3, etc. are acting simultaneously on a particle may be replaced by a single force, which will produce the same effect as produced by all the given forces.

Single force $\longrightarrow$ Resultant force

F_1, F_2 and F_3 $\longrightarrow$ Component forces

To find out the resultant force of a force system following methods are used.

(i) Analytical Method.

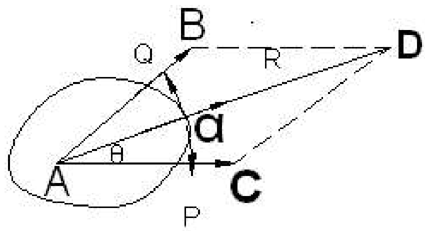
- Parallelogram Law.
- Triangle law of Forces.
- Methods of resolution & composition.

(ii) Graphical Method.

- Polygon Law of forces.

1.6.1. Parallelogram law of forces

"*If two forces, acting at a point be represented in magnitude and direction by the two adjacent sides of a parallelogram, then their resultant is represented in magnitude and direction by the diagonal of the parallelogram passing through that point*".



Consider two forces P and Q acting on a body at A as shown in fig1.4 (a). Let angel between the two forces be α. The resultant R of these two forces is obtained by the diagonal AD of the parallelogram ABDC, as shown in fig1.4 (a). The relationship between P, Q and R can be derived as follows:

Drop perpendicular from D and let it meet O extend at point C. In Δ CAD, side CA is parallel and equal to BD, i.e, it represents force P. The resultant R of P and Q is given by

$$R = AD = \sqrt{AO^2 + OD^2} = \sqrt{(AC + CO)^2 + OD^2}$$

But AC= P

CO= CD cosα= Qcosα

OD= CDsinα= Q sinα

$$R = \sqrt{(P + \mathrm{Q}\cos\alpha)^2 + (\mathrm{Q}\sin\alpha)^2}$$

$$\sqrt{P^2 + Q^2\cos^2\alpha + 2PQ\cos\alpha + Q^2\sin^2\alpha}$$

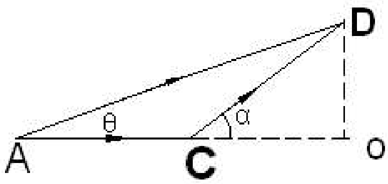
$$= \sqrt{P^2 + Q^2(\sin^2\alpha + \cos^2\alpha) + 2PQ\cos\alpha}$$

$$\sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$$

The inclination of the resultant R to the direction of force P is given by

$$\tan\theta = \frac{OD}{AO} = \frac{OD}{AC+CO} = \frac{Q\sin\alpha}{P+Q\cos\alpha}$$

$$\theta = \tan^{-1}\left[\frac{Q\sin\alpha}{P+Q\cos\alpha}\right]$$



Note: it is necessary that for the law of parallelogram for forces to be valid, one of the two forces should be along the x-axis. The forces P and Q may be in any direction as shown in Figure below.

If the angle between the forces is θ, then

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$$

The direction of the resultant will be

$$\theta = \tan^{-1}\frac{Q\sin\alpha}{P+Q\cos\alpha}$$

1.6.2. Triangle law

If the two forces acting simultaneously on a body are represented by the sides of a triangle taken in order, then their resultant is represented by the closing side of the triangle in the opposite order.

Remark: In the triangle of forces it must be carefully noted that the forces must be parallel to the sides of a triangle taken in order, i.e. taken the same way round.

1.6.3. Lami's theorem

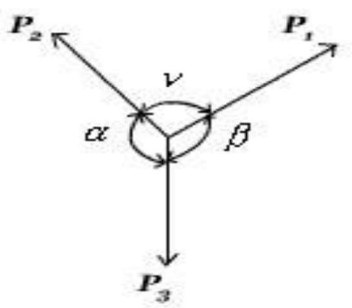


Fig. Coplanar and concurrent forces

If three forces coplanar and concurrent forces acting on a particle keep it in equilibrium, then each force is proportional to the sine of the angle between the other two and the constant of proportionality is the same.

For the system shown in the above figure,

$$\frac{P_1}{\sin\alpha} = \frac{P_2}{\sin\beta} = \frac{P_1}{\sin\gamma}$$

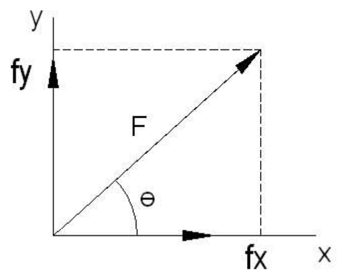
1.6.4. Resolution of forces

The replacement of a single force by several components which will be equivalent in action to the given force is called resolution of force. In particular, cases where the two components act at right angles to each other are called rectangular components.

Resolving a force into rectangular components:

A force F is to be resolved into two rectangular components along x and y axes. The components F_x and F_y are obtained using the parallelogram law,

if θ is the angle between F and x axis



F_x= F cosθ F_y= Sinθ

tanθ= F_y/F_x

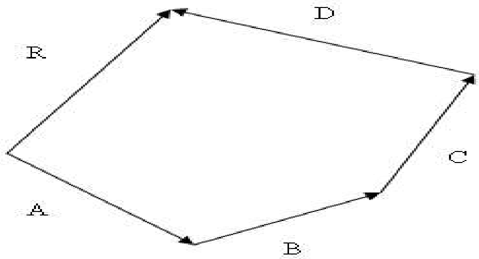
$$F = \sqrt{Fx^2 + Fy^2}$$

When a vertical force F is resolved into components tangential and perpendicular to an inclined plane, the component tangential to force is F_t= F sinα

Component vertical to force F_n= F cosα

1.6.5. Polygon Law of Forces

If a number of forces acting simultaneously on a particle be represented in magnitude and direction by the sides of a polygon taken in order, their resultant may be represented in magnitude and direction by the closing side of the polygon taken in opposite order.

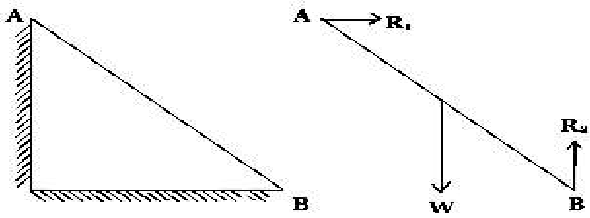


Where A, B, C and D are the forces and R = Resultant force.

1.7. Equation of equilibrium

A body is said to be in equilibrium, when it is at a state of rest or it will continue to move uniformly in a particular direction. In statics a body is at a state of rest under a system of forces, when R=0.

A particle is in equilibrium if it is stationary or it moves uniformly relative to an inertial frame of reference. A body is in equilibrium if all the particles that may be considered to comprise the body are in equilibrium. One can study the equilibrium of a part of the body by isolating the part for analysis. Such a body is called a free body. We make a free body diagram and show all the forces from the surrounding that act on the body. Such a diagram is called a free-body diagram. For example, consider a ladder resting against a smooth wall and floor. The free body diagram of ladder is shown in the right.



Three forces are acting on the ladder. Gravitational pull of the earth (weight), **W** of the ladder, reaction of the floor R_2 and reaction of the wall R_1. When a body is in equilibrium, the resultant of all forces acting on it is zero. Thus that resultant force **R** and the resultant couple $\mathbf{M}_{\mathbf{R}}$ are both zero, and we have the equilibrium equations

$$R = \sum F = 0 \quad \text{and} \quad M_R = \sum M = 0$$

These requirements are necessary and sufficient conditions.

Equilibrium of Coplanar Concurrent forces:

If all the forces in a system lie in a single plane and pass through a single point, then the system constitutes a coplanar concurrent force system. The resultant R of a system of forces is zero if and only if it's resolved components in two orthogonal directions x and y are zero. Moment about the common point of forces is obviously zero.

$$\sum F_x = 0 \qquad \sum F_y = 0$$

Condition is necessary:

Let R make an angle α with the x-axis. The sum of the resolved components of the forces is equal to the resolved component of resultant. Thus,

$$\sum F_x = R\cos\alpha = 0$$

$$\sum F_y = R\sin\alpha = 0$$

Condition is sufficient:

Squaring both equations and adding,

$$R^2\cos^2\alpha + R^2\sin^2\alpha = R^2 = 0$$

Hence, **R** will be zero.

It is not necessary that directions x and y should be orthogonal. Only the angle between them should not be 0^0 or 180^0. Necessity condition is obvious, because the resolved component of a zero force in any direction is zero.

Types of Equilibrium: There are two types of Equilibrium,

(i) Static Equilibrium:

- Velocity, V=0, a=0
- ΣF=0
- ΣM=0

(ii) Dynamic Equilibrium:

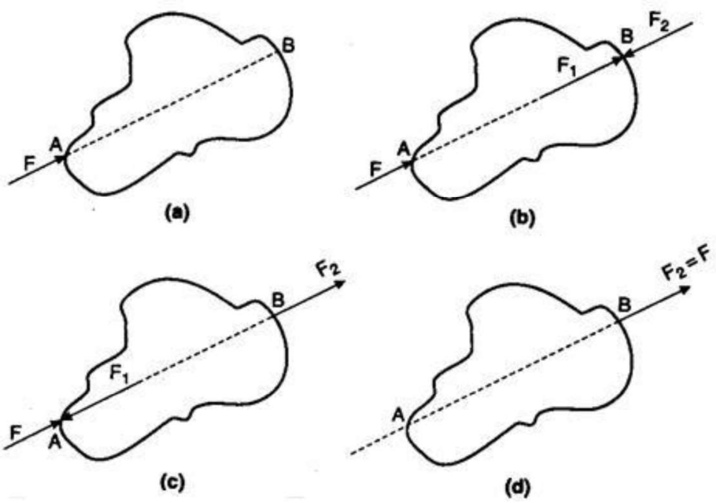
- Linear Velocity, V= Const.
- Angular Velocity= Const.

Equilibrium of statics is also classified as

- Stable equilibrium.
- Unstable equilibrium.
- Neutral equilibrium.

1.8. Principle of transmissibility of forces:

The state of rest or of motion of a rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting anywhere on the body along the line of action of the applied forces. In the following animation, two rigid blocks A and B are joined by a rigid rod. If the system is moving on a frictionless surface, the acceleration of the system in both the cases is given by,

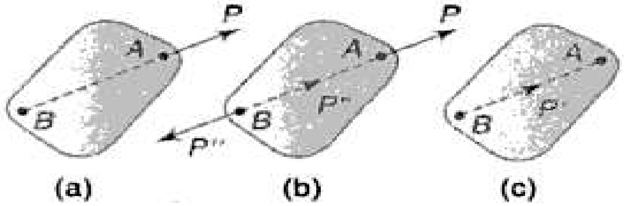


Acceleration=Applied force/total mass.

It is independent of the point of application

1.9. Law of superposition:

The action of a given system of forces on a rigid body will no way be changed if we add to or subtract from another system of forces in equilibrium.

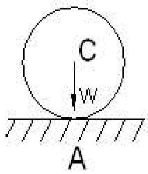


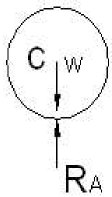
A rigid boy AB is under the action of force P applied at A acting along BA as shown in fig. If we add two equally opposite forces at point B which are in equilibrium, and collinear with P, then also the action of three forces on the body is identical with the action of single force P. Again if we remove the two equal and opposite forces P and P" from the system, then equilibrium of the system will not be altered. This statement process the theorem of transmissibility of a force.

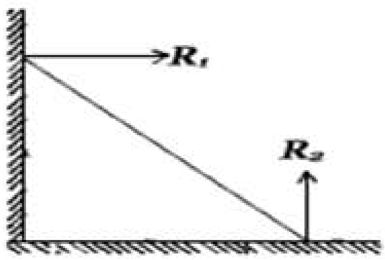
1.10. Law of action and reaction:

Very often we have to investigate the conditions of equilibrium of bodies that are not entirely free to move. Restriction to the free motion of a body in any direction is called constrain. A body that is not entirely free to move and is acted upon by some applied force (or forces) will, in general, exert pressures against its supports. These actions of a constrained body against its supports induce reactions from the supports on the body, and as the fourth principle of statics we take the following statement: Any pressure on a support causes an equal and opposite pressure from the support so that action and reaction are two equal and opposite forces.

These are self-adjusting forces developed by the other bodies, which come in contact with body under consideration. When two bodies come in contact, both apply equal and opposite force on each other. If the contact surface is smooth, the force will be in the normal direction of surface of contact. This is not true, if friction is present. For example, consider the frictionless contact of the following bodies: One ball kept on the other ball. In this case the contact forces will be along common normal.



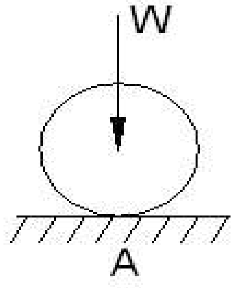


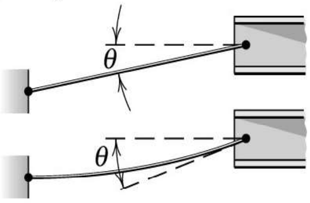
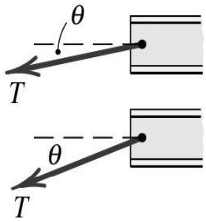
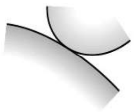
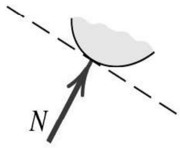
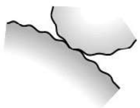
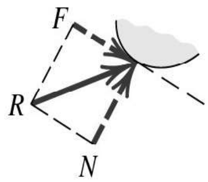
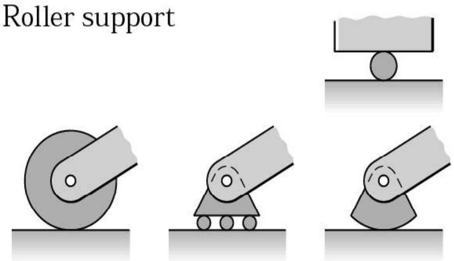
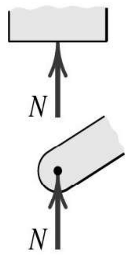
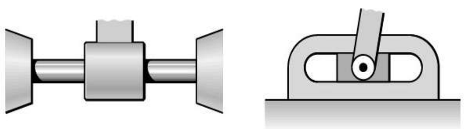
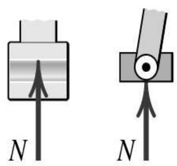


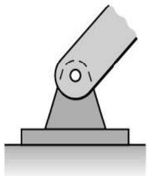
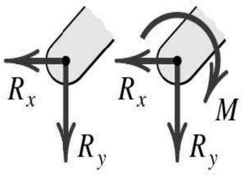
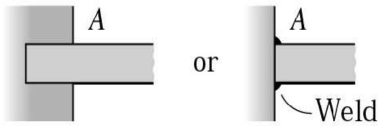
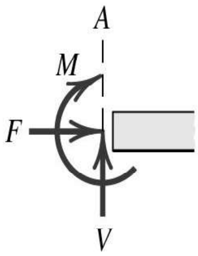
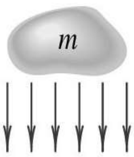
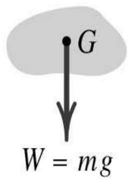
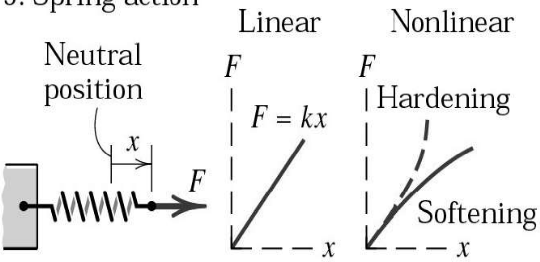
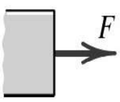
A ladder, in this case, the reaction of the wall will be in the horizontal direction and that of floor in the vertical direction. The pin joint can supply the reactive force in any direction. The fixed joint supplies reactive force as well as moment.

1.11. Constraints:

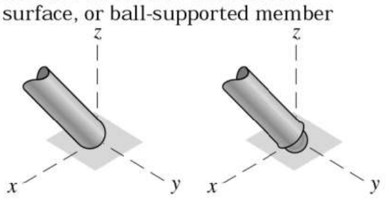
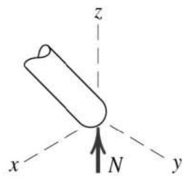
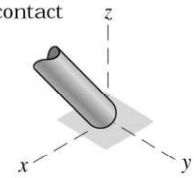
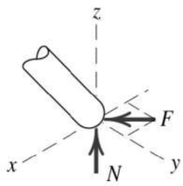
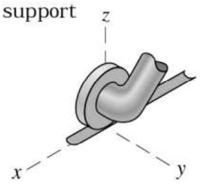
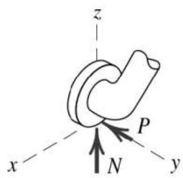
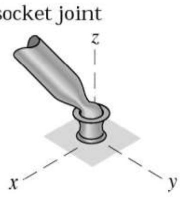
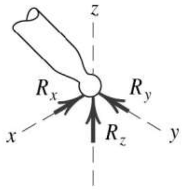
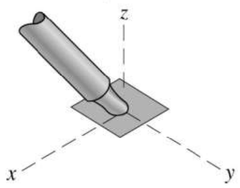
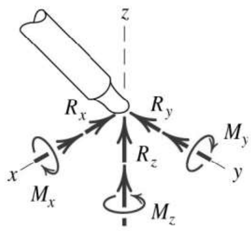
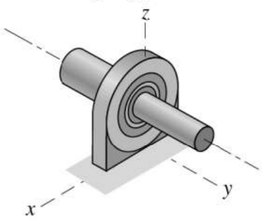
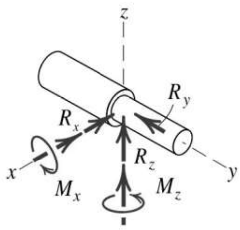
Restriction to free motion of a body in any direction is called constraint. e.g. A ball resting on a horizontal plane is free to move along the plane, but cannot move vertically downward direction as shown in fig. The ball exerts a vertical push against the surface of supporting plane at the point of contact A. The actions of a constrained body against its support induce reactions from the support on the body



MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
1. Flexible cable, belt, chain, or rope Weight of cable negligible Weight of cable not negligible 	 Force exerted by a flexible cable is always a tension away from the body in the direction of the cable.
2. Smooth surfaces 	 Contact force is compressive and is normal to the surface.
3. Rough surfaces 	 Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant contact force R.
4. Roller support 	 Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.
5. Freely sliding guide 	 Collar or slider free to move along smooth guides; can support force normal to guide only.

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS (*cont.*)	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
6. Pin connection 	 A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the axis; usually shown as two components R_x and R_y. A pin not free to turn may also support a couple M.
7. Built-in or fixed support 	 A built-in or fixed support is capable of supporting an axial force F, a transverse force V (shear force), and a couple M (bending moment) to prevent rotation.
8. Gravitational attraction 	 The resultant of gravitational attraction on all elements of a body of mass m is the weight $W = mg$ and acts toward the center of the earth through the center mass G.
9. Spring action 	 Spring force is tensile if spring is stretched and compressive if compressed. For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.

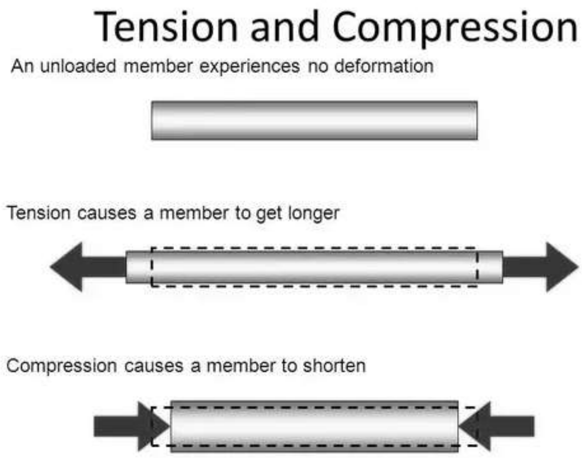
MODELING THE ACTION OF FORCES IN THREE-DIMENSIONAL ANALYSIS

Type of Contact and Force Origin	Action on Body to Be Isolated	
1. Member in contact with smooth surface, or ball-supported member 		Force must be normal to the surface and directed toward the member.
2. Member in contact with rough surface 		The possibility exists for a force F tangent to the surface (friction force) to act on the member, as well as a normal force N.
3. Roller or wheel support with lateral constraint 		A lateral force P exerted by the guide on the wheel can exist, in addition to the normal force N.
4. Ball-and-socket joint 		A ball-and-socket joint free to pivot about the center of the ball can support a force **R** with all three components.
5. Fixed connection (embedded or welded) 		In addition to three components of force, a fixed connection can support a couple **M** represented by its three components.
6. Thrust-bearing support 		Thrust bearing is capable of supporting axial force R_y as well as radial forces R_x and R_z. Couples M_x and M_z must, in some cases, be assumed zero in order to provide statical determinacy.

1.12. Tension & Compression:

Tension:

It is the axial force of resistance, which opposes the elongation of a member (rod or string). It is always away from the point of consideration & along the member. Strings are always in tensions.



Compression:

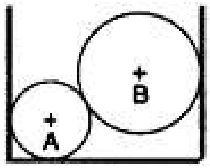
It is the axial force of resistance, which opposes the contraction of a member (rod), under external load. Direction of compressive resistance is always towards the point & along the members.

1.13. Free body diagram

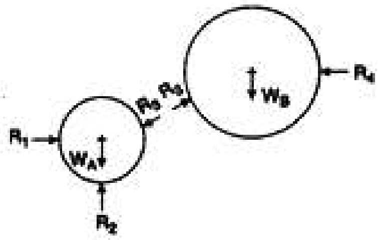
Free body diagram is a sketch of the isolated body, which shows the external forces on the body and the reactions exerted on it by removed elements.

General procedure for constructing a free body diagram:

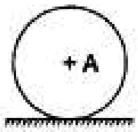
1. A sketch of the body is drawn, by removing the supporting surfaces.
2. Indicate on the sketch all the applied or active forces which tend to set the body in motion. e.g. weight, applied forces
3. Also indicate the reactive forces caused by supports preventing motion. Sense of unknown reaction should be assumed.
4. All relevant dimensions and angles, reference axes should be shown on sketch.



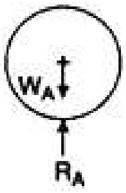
b(i) Two spheres in equilibrium



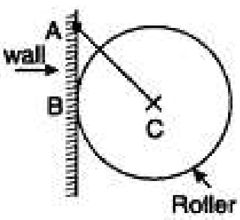
b(ii) Free body diagram



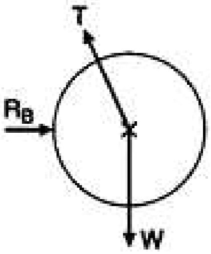
(c) Ball resting on a surface



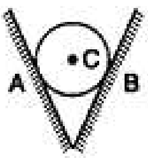
c (ii) Free body diagram



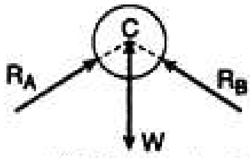
g(i) Hanging roller



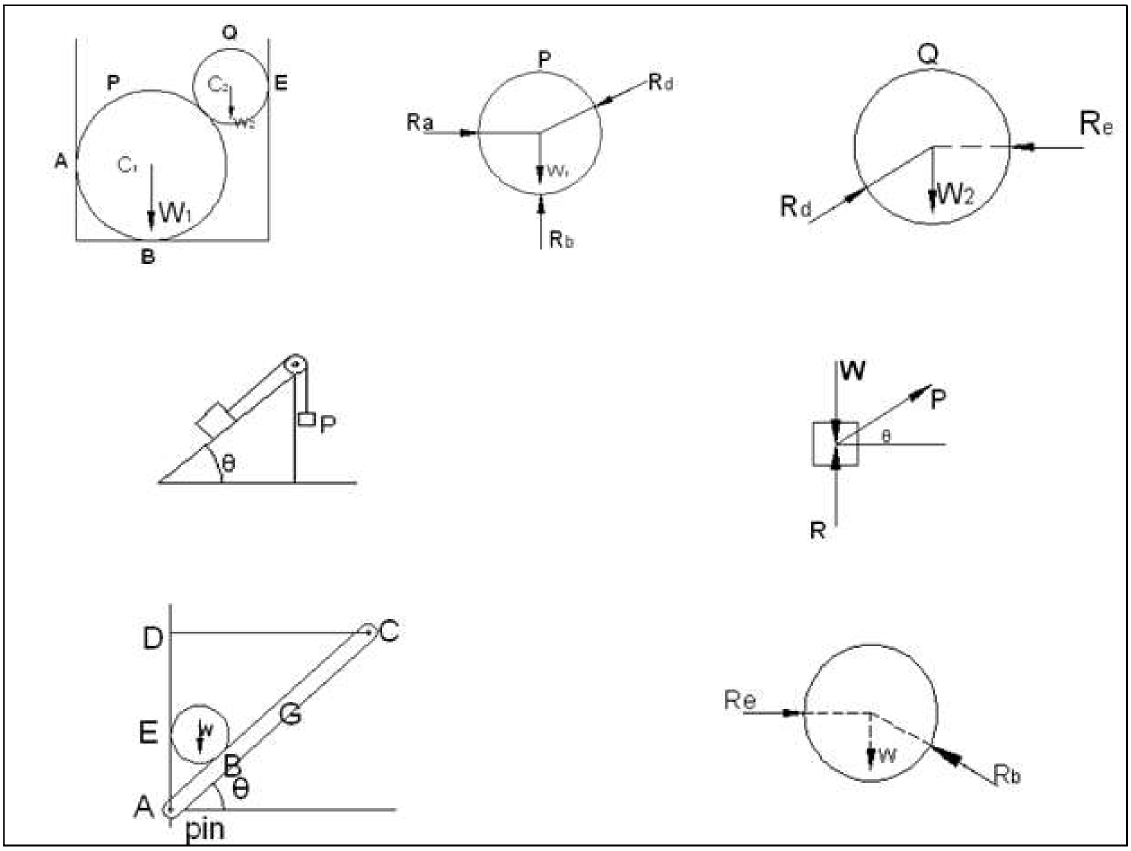
g(ii) Free body diagram



h(i) Sphere in groove

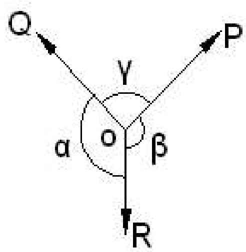


h(ii) Free body diagram



1.14. Lami's therorem

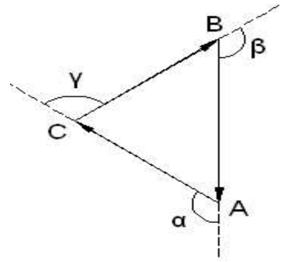
If three concurrent forces are acting on a body, kept in equilibrium then each force is proportional to the sine of the angle between the other two forces and the constant of proportionally is the same.



Let us consider forces P, Q and R acting at point O. Since the forces are in equilibrium, the triangle made by the forces is closed.

Drawing the triangle of forces ΔABC corresponding to the forces P and R acting at O as shown in fig.

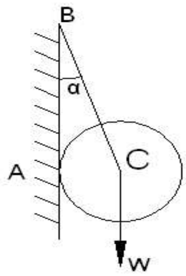
The angles are



$$\angle BAC = \pi - \alpha$$
$$\angle CBA = \pi - \beta$$
$$\angle ACB = \pi - \gamma$$

From sine rule for triangle

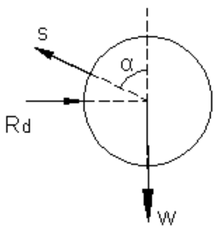
$$\frac{P}{Sin(\pi-\alpha)} = \frac{Q}{Sin(\pi-\beta)} = \frac{R}{Sin(\pi-\gamma)}$$

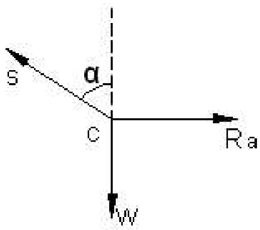


$$\frac{\mathrm{P}}{\mathrm{Sin}\alpha} = \frac{\mathrm{Q}}{\mathrm{Sin}\beta} = \frac{\mathrm{R}}{\mathrm{Sin}\gamma} \qquad \because \mathrm{Sin}(\pi-\alpha)$$

$$\mathrm{Sin}(\pi-\beta)$$
$$\mathrm{Sin}(\pi-\gamma)$$

e.g. a ball supported in a vertical plane by a string and a smooth wall.





Drawing the F.B.D. of the ball the three forces Ra, S and W will make force triangle.

Lami's theorem is applicable to three forces

$$\frac{\mathrm{W}}{\mathrm{Sin}(90+\alpha)} = \frac{\mathrm{Ra}}{\mathrm{Sin}(180-\alpha)} = \frac{\mathrm{S}}{\mathrm{Sin}90^\circ}$$

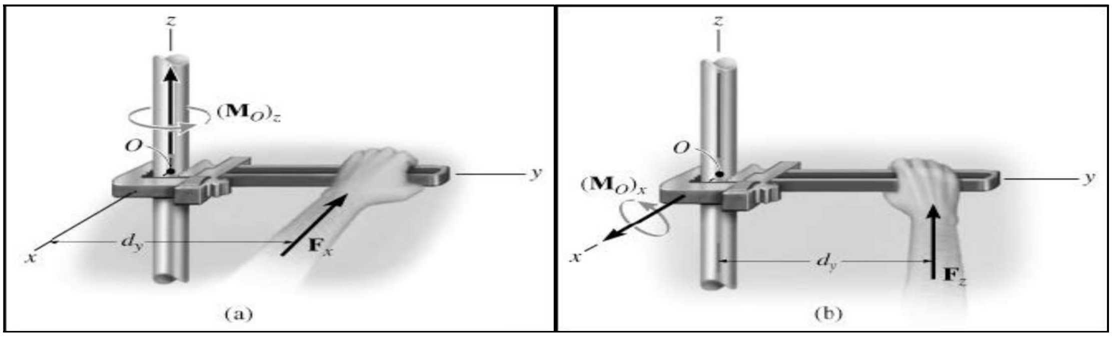
$$\Rightarrow \mathrm{S} = \mathrm{WSec}\alpha$$

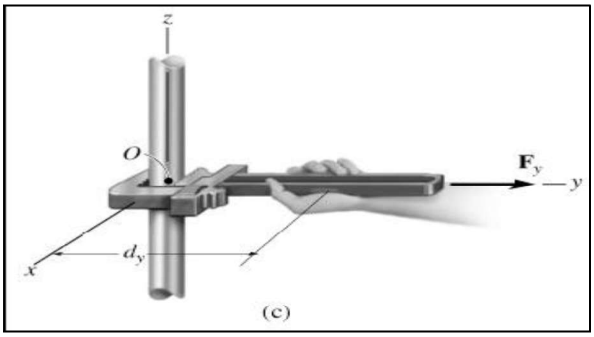
$$\Rightarrow \mathrm{R_a} = \mathrm{WTan}\alpha$$

1.15. Moment of a Force

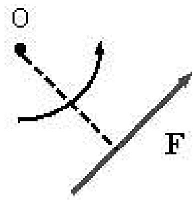
The moment of a force about a point O is the cross product of ***r*** and ***F***, where ***r*** is the position vector of any point on the line of action of force with respect to O.

$$M = r \times F$$

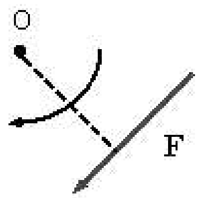




Just as force has a tendency to translate the body, moment has a tendency to rotate the body about the point. Our sign convention will be as follows. If the force has tendency to rotate the body in counterclockwise sense, moment is considered +ve. If the force has tendency to rotate the body in clockwise sense, moment is considered -ve.



CCW-out of the page



CW-into the page

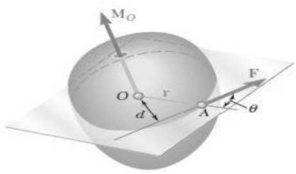
A convenient way to find out the direction of moment is as follows. Imagine to stand on the line of action of force , facing yourself in the direction of arrow. If the point O (about which moment is desired) is towards your left, moment is positive. If the point is towards your right, moment is negative.

Moment of a force about a point

Let us now consider a force **F** acting on a rigid body fig2.39. As we know, the force **F** is represented by a vector which defines its magnitude and direction. However, the effect of the force on the rigid body depends also upon its point of application *A*. The position of *A* can be conveniently defined by the vector **r** which joins the fixed reference point *O* with *A* ; this vector is known as the *position vector* of *A*. The position vector **r** and the force **F** define the plane.

We will define the *moment of* **F** *about O* as the vector product of **r** and **F:**

$$\mathbf{M_o = r \times F}$$



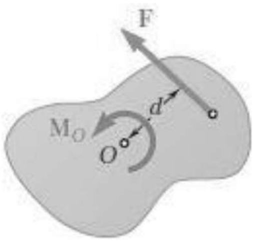
The moment $\mathbf{M}_O$ must be perpendicular to the plane containing *O* and the force **F**.

The sense of $\mathbf{M}_O$ is defined by the sense of the rotation which will bring the vector **r** in line with the vector **F**; this rotation will be observed as *counterclockwise* by an observer located at the tip of $\mathbf{M}_O$. Another way of defining the sense of $\mathbf{M}_O$ is furnished by a variation of the right-hand rule: Close your right hand and hold it so that your fingers are curled in the sense of the rotation that **F** would impart to the rigid body about a fixed axis directed along the line of action of $\mathbf{M}_O$; your thumb will indicate the sense of the moment $\mathbf{M}_O$ shown in fig.2.39. Finally, denoting by u the angle between the lines of action of the position vector **r** and the force **F**, we find that the magnitude of the moment of **F** about *O* is $\mathbf{M}_O = \mathbf{r\ F\ Sin\ \Theta = F\ d}$ where *d* represents the perpendicular distance from *O* to the line of action of **F** . Since the tendency of a force **F** to make a rigid body rotate about a fixed axis perpendicular to the force depends upon the distance of **F** from that axis as well as upon the magnitude of **F**, we note that the magnitude of $\mathbf{M}_O$ measures the tendency of the force **F** to make the rigid body rotate about a fixed axis directed along $\mathbf{M}_O$.

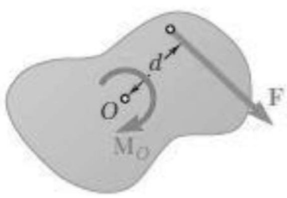
In the SI system of units, where a force is expressed in Newton (N) and a distance in meter (m), the moment of a force is expressed in Newton-meter (N. m).

We can observe that although the moment $\mathbf{M}_O$ of a force about a point depends upon the magnitude, the line of action, and the sense of the force, it does *not* depend upon the actual position of the point of application of the force along its line of action. Conversely, the moment $\mathbf{M}_O$ of a force **F** does not characterize the position of the point of application of **F**.

Problems Involving Only Two Dimensions: Many applications deal with two-dimensional structures, i.e., structures which have length and breadth but only negligible depth and which are subjected to forces contained in the plane of the structure. Two-dimensional structures and the forces acting on them can be readily represented on a sheet of paper or on a blackboard. Their analysis is therefore considerably simpler than that of three-dimensional structures and forces. The sense of the moment of **F** about *O* in **fig 2.40 (*a)*** as counter clockwise, and in **fig 2.40 (*b)*** as clockwise



(a) $M_o = +F\ d$



(b) $M_o = -F\ d$

Since the moment of a force **F** acting in the plane of the figure must be perpendicular to that plane, we need only specify the magnitude and the sense of the moment of **F** about O. This can be done by assigning to the magnitude M_O of the moment a positive or negative sign according to whether the vector $\mathbf{M}_O$ points out of or into the paper.

Varignon's Theorem: Moment of a force about any point is equal to the sum of the moments of the components of that force about the same point. To prove this theorem, consider the force ***R*** acting in the plane of the body shown in Fig. 1.4. The forces ***P*** and ***Q*** represent any two nonrectangular components of ***R*.** The moment of ***R*** about point O is $\boldsymbol{M_0} = \boldsymbol{r} \times \boldsymbol{R}$

Because $\boldsymbol{R} = \boldsymbol{P}+\boldsymbol{Q}$, we may write

$\boldsymbol{r} \times \boldsymbol{R} = \boldsymbol{r} \times (\boldsymbol{P} + \boldsymbol{Q})$

Using the distributive law for cross products, we have

$\boldsymbol{M_0} = \boldsymbol{r} \times \boldsymbol{R} = \boldsymbol{r} \times \boldsymbol{P} + \boldsymbol{r} \times \boldsymbol{Q}$

which says that the moment of $\boldsymbol{R}$ about $\boldsymbol{O}$ equals the sum of the moments about $\boldsymbol{O}$ of its components $\boldsymbol{P}$ and $\boldsymbol{Q}$. This proves the theorem.

Varignon's theorem need not be restricted to the case of two components, but it applies equally well to three or more. where we take the clockwise moment sense to be positive.

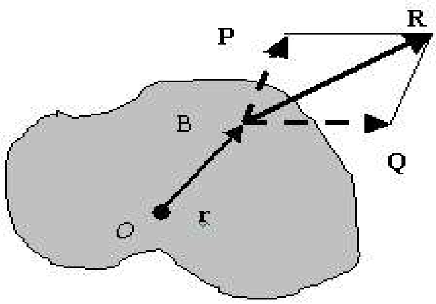


Figure: Illustrating Varignon's theorem

1.16. Friction

Friction is the force distribution at the surface of contact between two bodies that prevents or impedes sliding motion of one body relative to the other. This force distribution is tangent to the contact surface and has, for the body under consideration, a direction at every point in the contact surface that is in opposition to the possible or existing slipping motion of the body at that point.

See this animation. A force is applied to body sliding on the floor. After sometime, the force is removed, but the body keeps moving, since it has already attained the velocity. However, after some time the body comes to rest. It is the frictional force, which causes body to stop. If the frictional force were not there, body would have kept moving.

If we want that body should keep moving at a constant speed, some force has to be applied to overcome the frictional force. In the following animation, body is moving with constant velocity on a rough surface. Note that, force is doing some work. This work, is however is neither increasing the kinetic energy nor the potential energy of the block. Instead, it is wasted as heat. In absence of friction no wastage of work. Friction has very positive benefits too! A man moving on a slope got tired. For taking rest, he keeps the pot on the slope. Had friction not there, pot would not have been able to rest on the slope. It would have slipped down. Without friction, we cannot place anything on the floor. Even an ant will be able to push away the object. In the following animation, an ant is pushing a heavy box, which is resting on a frictionless surface. For stopping the automobile, we depend on the friction. Friction is needed for some manufacturing processes. One such process is falt rolling, in which a sheet is passed between two counter-rotating cylinders to reduces its thickness. Sheet is drawn by the friction force.

Types of friction

(1) Dry friction

Dry friction occurs when the unlubricated (rough) surfaces of two solids are in contact under a condition of sliding or a tendency to slide. A friction force tangential to the surfaces of contact occurs both during the interval leading up to impending slippage and while slippage takes place. The direction of this friction force always opposes the motion or impending motion. This type of friction is also called Coulomb friction.

(2) Fluid friction

Fluid friction occurs when adjacent layers in a fluid (liquid or gas) are moving at different velocities. This motion causes frictional forces between fluid elements, and these forces depend on the relative velocity between layers. When there is relative velocity between layers, there is no fluid friction. Fluid friction depends on the velocity gradients within the fluid and on the viscosity of the fluid.

(3) Internal friction

Internal friction occurs in all solid materials subjected to cycle loading.

Dry friction

Consider a solid block of mass *m* resting on a horizontal surface as shown. Assume that the contacting surfaces are rough. As we gradually increase the load, the block remains static till the load reaches a threshold value. Since the forces in the *x*-direction have to be balanced, it is apparent that as the magnitude of *P* increases from zero, the friction force also increases. Friction force, is thus, self-adjusting. However, the friction force cannot increase beyond a limit. Thus there is a limiting value of friction. The maximum value of friction force, which comes into play, when the motion is impending, is known as limiting friction. When the applied force is less than the limiting friction, the body remains at rest and such frictional force is called static friction, which may have any value between zero and the limiting friction. If the value of the applied force exceeds the limiting friction, the body starts moving over the other body and the frictional resistance experienced by the body while moving is known as Dynamic friction. dynamic friction is found to be less than limiting friction. See the following animation to understand the phenomenon of dry friction

It is experimentally found that the magnitude of limiting friction bears a constant ratio to the normal reaction between the two surfaces and this ratio is called coefficient of Friction.

Coefficient of friction = F/N, where *F* is limiting friction and *N* , the normal reaction between the contact surfaces.

Laws of dry friction

(i) Force of friction always act opposite to the movement of body.

(ii) The friction force developed at any interface is directly proportional to the normal reaction transmitted across the surface of contact.

$$F \propto N$$

$$F = \mu N$$

F - Force required to start sliding or is equal to limiting friction.

N - Normal reaction between the surfaces.

μ- coefficient of static friction.

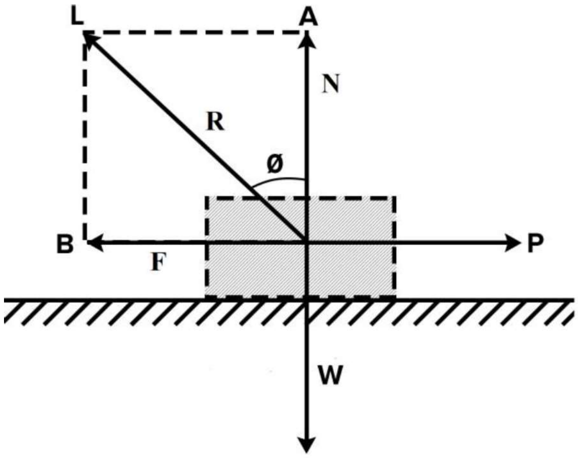
F = μN is valid only at the time of impending motion.

After the motion stands μ is called the coefficient of kinetic friction.

(iii) The total friction that can be developed is independent of the area in contact and dependent on roughness of the surface.

Angle of static friction

Angle made by the resultant of normal reaction and limiting friction with the normal reaction is called the angle of friction. The coefficient of static friction is equal to the tangent of the angle of friction. Consider the block on the following surface.



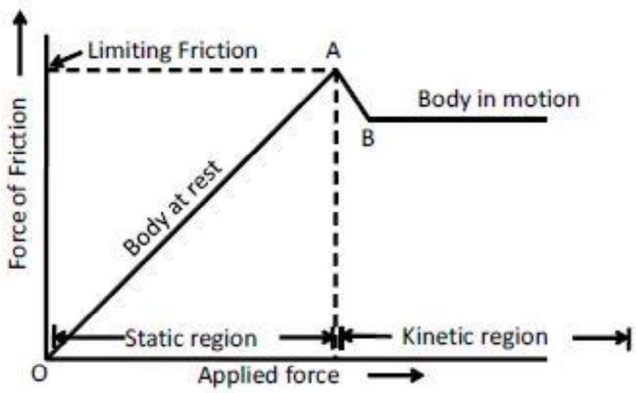
The free body diagram is shown. The direction of resultant *R* measured from the direction of *N* is specified by tan Ø =*F*/*N*. When the friction force reaches its limiting static value F_{max} , the angle a reaches a maximum value Ø. Thus

$$\tan Ø = \mu_s$$

The angle Ø is called the angle of static friction.

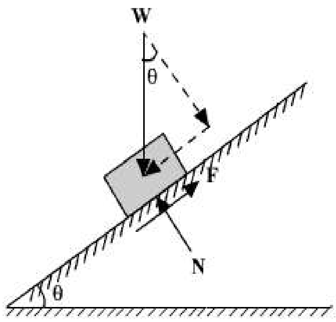
Limiting friction

Consider a body of mass μ having the corresponding weight W resting on a horizontal surface. Let a continuously increasing force be applied as shown in figure. The force P will be opposed or resisted by frictional force F. The F shall go on increasing to balance the increasing applied force P and the body shall remain rest. Then there comes a limit beyond which the frictional force cannot increase and the body begins to move. The frictional force at this instant is called limiting friction. The limiting friction is therefore is the maximum frictional force exerted at the time of impending motion, that is when the motion is about to begin.



The frictional force between two moving surfaces is called kinetic or dynamic friction.

Angle of Repose



The maximum inclination of the plane on which a body, free from external forces, experiences repose (sleep) is called **Angle of Repose**.

Now consider the equilibrium of the block shown above. Since the surface of contact is not smooth, not only normal reaction, but frictional force also develops. Since the body tends to slide downward, the frictional resistance will be up the plane.

$\sum$forces normal to the plane =0, gives

$$N = W\cos\alpha$$

$\sum$forces normal to the plane =0, gives

$$F = W\sin\alpha$$

Dividing equation (2) by equation (1), we get

$$\tan\theta = \frac{F}{N}$$

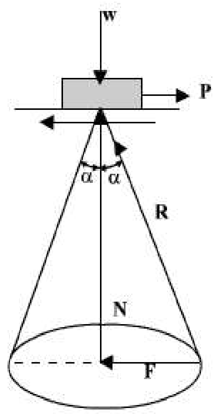
If N is the value of normal force when motion is impending, frictional force will be μN and hence

$$\tan\theta = \frac{F}{N}$$
$$= \mu$$
$$= \tan\alpha$$
$$\text{or,} \quad \theta = \alpha$$

Hence, to avoid free sliding, the inclination angle should be less than the friction angle.

Cone of friction

When a body is having impending motion in the direction of *P* the frictional force will be the limiting friction and the resultant reaction R will make limiting friction angle α with the normal as shown in the following figure. If the body is having impending motion in some other direction, again the resultant reaction makes limiting frictional angle α with the normal in that direction. Thus, when the direction of force *P* is gradually changed $360^{0,}$ the resultant R generates a right circular cone with semi central angle equal to α.



If the resultant reaction is on the surface of this inverted right circular cone whose semi-central angle is limiting frictional angle α, the motion of the body is impending. If the resultant is within this cone the body is stationary. This inverted cone with semi-central angle equal to limiting angle α is called Cone of Friction.

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